Sahaj Bhadja

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Summary

Forward Deployed Engineer with 2+ years building and shipping **generative AI** workflows and apps with **Claude**, GPT-4o, and **RAG**, translating client processes into automations that delivered measurable gains like 60% faster turnaround.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Web Technologies, Database Systems, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Retrieval, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Engineer *Los Angeles, CA*
- Shipped a multi-tenant **AI education platform** on Azure SQL and Blob Storage for 500+ users, designing secure row-level data isolation and cutting dashboard latency from 1.2s to 350ms.
- Built fault-tolerant **LLM workflows** with FastAPI background workers and explicit job states, processing 1K+ AI jobs at 3x throughput while automated integrations ran 5K+ monthly jobs at 99%+ reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*
- Built a **generative AI** grading workflow that auto-generates rubrics, scores text and handwritten answers with GPT-4o, Claude, and Qwen-VL, and returns feedback, cutting turnaround 60% for 150+ instructors.
- Designed **prompt chains** and model fallbacks for OCR reliability, tracked per-call token cost to keep usage efficient, and scaled the workflow to 10K+ daily requests at 90%+ grading consistency.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*
- Automated cross-platform data-sharing **workflows** that cut manual data entry 35%, working with Microsoft and GitHub **stakeholders** in Agile sprints from requirement analysis through deployment and production support.
- Led a team redesigning ETL workflows on AWS Glue and S3, raising **pipeline efficiency** 68%, and automated nightly build, test, and reporting jobs that emailed coverage summaries and failure logs.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*
- Used Google Analytics on feature usage, traffic sources, and drop-off points to drive data-backed product changes that cut resource wastage 42% and raised CRM **lead conversions** 15%.
- Built a MERN resource-management CRM with eight microservices for resource allocation, billing, lead management, and notifications, syncing 2M+ records across locations through REST APIs and real-time **WebSockets**.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*
- Gathered **client requirements**, mapped 12+ integration points, and presented the end-to-end architecture for sign-off, then delivered Python booking, pricing, and payment APIs with Firebase real-time status sync.
- Automated refund, dispute, and chargeback handling with payment **webhooks**, cutting manual work 70%, and built hourly Databricks analytics on peak hours, service demand, and revenue for client reporting.

Skills

GenAI & Agents: Claude, GPT-4o, LLaMA-2, Prompt Engineering, Prompt Chaining, RAG, Embeddings, OCR, LLM Orchestration, Multi-Model Routing
AI-Assisted Building: Claude Code, Rapid Prototyping, Workflow Automation, Webhooks, REST API Integrations
Engineering: Python, TypeScript, JavaScript, SQL, React, Next.js, Node.js, FastAPI, Flask
Data & Cloud: PostgreSQL, Supabase, MongoDB, Firebase, Databricks, Google Analytics, AWS, Azure, GCP, Vercel, Docker
Client & Delivery: Requirements Gathering, Client Presentations, Stakeholder Management, Technical Documentation, Impact Measurement, Agile/Scrum

Projects

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, Convex, Clerk, Shadcn*
- Built and deployed an AI document assistant with LLaMA-2 **RAG**, real-time summarization, and Convex collaboration, cutting document review time 60% for 200+ users at 95% query accuracy.

PiXMod Live | *Next.js, SAM, Stable Diffusion, CLIP, MongoDB, Stripe*
- Shipped a **generative AI** SaaS app in Next.js with SAM segmentation, Stable Diffusion inpainting, and CLIP search over 100K+ images, adding Stripe subscriptions that lifted premium conversion 18%.

Chat2SQL Code | *Java, TypeScript, PostgreSQL, Snowflake, Docker*
- Built an AI-assisted SQL and reporting workspace that lets users query 10+ databases in plain language, generating schema-aware SQL at 82% accuracy and **AI-generated dashboards** for faster analysis.